

Simulating viral disease using Monte Carlo simulations

Results-

1. I developed the simulation code in python to plot in one graph the average number of students susceptible, infected and affected in total (removed) for the period of 40 days.
2. With 1% initial immunity and a 20% daily recovery chance, there is a rapid spike in infections that eventually burns out as the recovered population grows (herd immunity).
3. By tweaking the parameters (increasing initial immunity or speeding up recovery), the simulation visually demonstrates how the infection peak lowers and delays, reducing the immediate burden on the population
4. Endemic State (S-I-S): -A modified version of the code shows what happens if immunity isn't permanent (like the common cold). Instead of the disease dying out, the population enters an endless loop, reaching a steady "endemic" state.

Summary of results: -

By using python as a programming language and adopting Monte Carlo simulation technique, the model accurately recreated how an outbreak happens. I was able to plot a massive, fast spike in sick students that eventually faded away on its own once enough the infected became naturally immune. The code allowed us to test different situations. It clearly proved that starting with more immune people (like having a vaccinated population) successfully "flattens the curve," keeping the peak number of sick people low. On the flip side, if the virus takes a long time to recover from, the peak number of infections gets much, much worse.

Finally, I changed the rules of the code so that recovered students could catch the virus a second time (just like the common cold). It entered a never-ending loop where a steady number of people were always sick